Building Interoperable Location-based Augmented Reality with the Open AR Cloud

Gábor Sörös
Nokia Bell Labs
Budapest, Hungary
Open AR Cloud Europe gUG
Gräfelfing, Germany
gabor.soros@nokia-bell-labs.com

James Jackson
Open AR Cloud
Portland, Oregon, USA
james.jackson@openarcloud.org

Mikel Salazar
Institute for Energy Technology
Halden, Norway
Open AR Cloud Europe gUG
Gräfelfing, Germany
mikel.salazar@ife.no

Christine Perey
PEREY Research & Consulting
Montreux, Switzerland
Open AR Cloud Europe gUG
Gräfelfing, Germany
cperey@perey.com

Ali C. Hantal
Open AR Cloud
Portland, Oregon, USA
ali.hantal@openarcloud.org

Alina Kadlubsky
Open AR Cloud Europe gUG
Gräfelfing, Germany
communication@openarcloud.org



Figure 1: Our open and scalable AR cloud platform enables large-scale indoor and outdoor AR experiences. Our WebXR-based client application supports a wide variety of geo-anchored digital contents from 2D icons and 3D models over point of interest descriptors towards GeoPose streams of robotic agents. In this tutorial, participants will learn how to set up their own testbeds for location-based AR research projects.

Abstract

This tutorial provides an overview of location-based augmented reality and the spatial web, their use cases, the ecosystem, and challenges related to the lack of interoperability between proprietary solutions. The Open AR Cloud Association promotes an open spatial web and develops open-source components to achieve this goal. Participants will learn about these components, their current state, future steps, and gain access to the toolset and proof-of-concept use cases to to build their own location-based AR experiences anchored to the real world.

CCS Concepts

• **Human-centered computing** → **Ubiquitous and mobile computing systems and tools**; **Mixed / augmented reality**; *Collaborative and social computing systems and tools*; • **Information systems** → *Global positioning systems*; • **Networks** → *Location based services.*

Keywords

Mixed/augmented reality, Ubiquitous and mobile computing systems and tools, Location based services

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1 Introduction

Technologies like geospatial augmented reality (AR), the AR cloud, the spatial web, and spatial computing in general enable persistent, location-specific digital content that can be discovered and consumed in context at a specific physical location. The virtual content can vary from simple text and URLs over 2D icons, videos to 3D assets (static/dynamic) and 3D sounds, but also IoT sensor sensor data injected into complex mixed-reality experiences.

Location-based digital contents are applicable in a broad range of use cases in various industries such as maintenance, construction, transportation, urban planning, cultural heritage, gaming, healthcare marketing, manufacturing, entertainment and many more.

The technological enablers of a spatial web include include 3D mapping, localization with 6 degrees of freedom, content discovery, reality modeling, and the rendering of personalized content to users in AR sessions. Today, these technologies are provided in multiple vertically integrated proprietary platforms. To be widely adopted, services for an open and interoperable spatial web will need to continuously span the globe and provide value to diverse user segments and interests. A distributed architecture in which interoperable services are decoupled from one another, and from the XR client a user chooses, will permit experts to focus on research and development of unique new services and a great variety of service combinations. Our vision is for spatial web services to be configured dynamically, discoverable and delivered in context and in alignment with user interests or needs.

2 The Open AR Cloud

The global non-profit Open AR Cloud association (OARC) set as its mission to promote the adoption of interoperable spatial computing technologies to connect the digital and physical worlds. Over the past 5 years, the OARC community created open and interoperable components of an Open Spatial Computing Platform (OSCP), also published in [3–5]. Components of the OSCP are under constant development and are available on github[1] with a permissive license.

A very brief overview of the platform is illustrated in Figure 2. AR-capable clients (smartphones or smartglasses) exchange information with the platform services via open APIs. As the first step, a service discovery mechanism allows clients to find spatial services in their vicinity. In the second step, precise 6DoF pose estimation and continuous pose tracking is performed via camera-based positioning. The clients are brought into a common geospatial reference frame. In the third step, given the client's geopose, the platform helps to discover digital contents and points of interest available to the client at the current location. Optionally, reality modeling services can provide machine-readable digital models of the scene which are served to the client. These services together augment human capabilities by retrieving and rendering the most relevant information in situ.

3 Tutorial at Ubicomp 2025

Participants of this tutorial will explore the use of the Open Spatial Computing Platform (OSCP) for interoperable publishing, discovery and consumption of geo-anchored mixed reality content. Our solution is a research framework now, but it can enable a wide range of use cases and applications.

It is important to note that the tutorial is not exclusively about the OSCP platform, but instead we will introduce the general concepts, the architecture and required technological components of a spatial web, and we will walk through the existing OSCP components only as examples. Some components already have multiple different realizations, and comparing them serves as a reference for the audience for potentially creating new ones and integrating their own location-based services.

[1] https://github.com/OpenArCloud

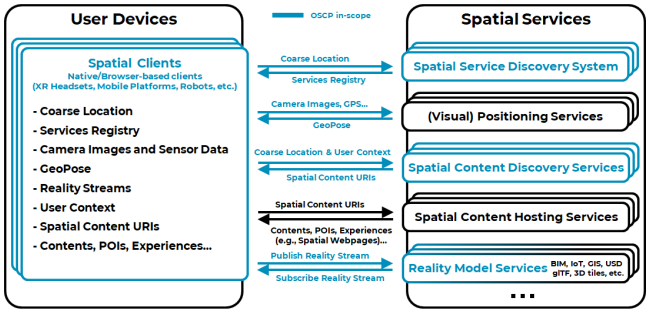


Figure 2: Overview of the interactions between clients and services of the platform: service discovery, localization, reality modeling, content discovery, content retrieval, etc.

In the past, there were already two tutorials about the OARC at IEEE VR 2021 [2] and at IEEE ISMAR 2021 [1] and since then a lot of developments happened and we would like to introduce the platform to the Ubicomp and ISWC community as a research tool for building location-based experiences.

We plan presentation sessions and live coding sessions where participants can learn how they can map their own space, register services and contents to geolocations, and create new experiences. The website of the tutorial is available at https://www.openarcloud.org/ubicomp.

3.1 Intended audience

The target audience for this tutorial includes academics and researchers as well as industry professionals working on ubiquitous computing, location-based services, mixed reality, computer vision, and related fields who are interested in the intersection of localization, (geo-)spatial information systems, as well as immersive technologies, AI assistants, smartglasses, and metaverse projects. We believe that these fields largely overlap with the interests of the Ubicomp/ISWC community.

All levels of expertise are welcome and encouraged to participate. This is an introductory tutorial, open to all with an interest in open spatial computing. Basic familiarity with digital content creation, coding for the web (WebXR) or 3D engines (Unity) is an advantage but not required. We expect 20-40 participants.

3.2 Expected value for the audience

Participants will gain knowledge about the technology stack required for interoperable publishing, discovery and consumption of content anchored to a 'digital-twin' of a real-world location and how it relates to other technologies such as 5G, cloud/edge computing and smart sensors. Instructors will introduce the participants to some of the difficulties of creating open source solutions, but also the critical importance of having an interoperable Spatial Web. By fostering collaboration among researchers and industry professionals, the workshop aims to bridge existing knowledge gaps and set the foundation for future innovations.

3.3 Requirements

The tutorial is planned for half day duration. While local participation is preferred, the tutorial can be also joined remotely (via Teams or Zoom), if the conference allows.

At the location, stable lighting conditions and a relatively static environment are important for robust visual localization. We will need to map the venue in advance to be able to provide localization service for the live demonstrations.

Regarding hardware, we only require a projector and presenter podium onsite. Participants are expected to bring their own laptop and Android mobile phone (iOS currently does not support WebAR, unfortunately).

4 Contents

The tutorial includes a general introduction to spatial computing concepts, the technology stack required for persistent mixed reality content, guided demonstrations of existing platform components and tools, mapping and localization at the conference venue, and a hands-on coding session of a client application.

4.1 Planned contents and schedule

During this tutorial, participants will gain knowledge about the technology stack of a spatial web required for interoperable publishing, discovery and consumption of content anchored to a 'digital twin' of real-world locations and how it relates to other technologies such as 5G, edge computing and smart sensors.

The schedule is planned for 4 hours (including a coffee break) with the following sessions:

- Spatial web basics (15 mins): Vision, state of the art, applications, challenges.
- Platform components (30 mins): Overview of the Open AR Cloud platform, spatial search and discovery, content authoring, content placement, positioning basics.
- GeoPose (15 mins): Geo reference systems, OGC GeoPose standard for universal shared reference frame between different AR clouds, 3D visualization, pose streams.
- Visual positioning (45 mins): Visual positioning systems (VPS), technical background, why open source solutions and interoperability between these systems need to be established. Building a point-cloud 'digital-twin' and 3D map of a real-world location. Introduction to the solutions of Immersal and Augmented City, then an open GeoPoseProtocol, and Nokia's Open VPS implementation.
- Coffee break (20 mins)
- Reality modeling (20 mins): The need fo machine-readable digital representations of the world, reality modeling language, use cases.
- Client applications (35 mins): We have a WebXR-based and a Unity-based client, and in this hands-on tutorial we focus on WebXR, which is free for everyone and can run in a browser without installation. We show the client architecture and how they can be modified and extended for own experiments.
- Content and discovery (15 mins): Available content types, content authoring and placement, point of interest (PoI) service, OGC PoI standard, etc. Demonstration of interoperable content placement between a Unity client (MyGeoVerse by XR-Masters) and a WebXR client (spARcl by OARC).
- Advanced features (15 mins): IoT sensor streams, 3D sounds, point clouds, photorealistic models, message brokers.
- Hands-on tutorial (30 mins): We plan to walk through the design and implementation of a location-based game, for example a collaborative virtual block world builder.
- Outlook and wrap-up (10 mins)

Over the course of the tutorial, we will gradually build up a showcase (similar to Fig. 1) including the geo-localization of users via VPS, the discovery of local PoIs, the creation of new objects, and multi-user interaction in a WebXR client.

4.2 Speakers

Besides the organizers from Open AR Cloud[2], we plan to include external experts from standards organizations like the Open Geospatial Consortium[3] and the Metaverse Standards Forum[4] as well as from companies like Immersal (FI)[5], Augmented City (IT)[6], XR-Masters (US) [7], etc. who will introduce their OSCP-compliant solutions for visual positioning and AR content placement. We are aiming for a gender-balanced speaker participation and a mix of both technical and humanities, industrial and academic backgrounds.

4.3 Hackathon and demonstrations

We plan to map the outdoor (and potentially indoor) space at the conference venue in advance and enable our platform at the area with a demonstration. The participants can experience interoperable GeoPose-compliant localization, standard Points of Interest (POIs) discovery, and AR content persistence and sharing across a native mobile app and a WebXR client. During and after the tutorial, interested participants, who want to build their own demo, are offered personal consultation hours throughout the conference week from the organizers and potentially even afterwards from the Open AR Cloud community.

5 Organizers

The organizer team consists of experts with diverse backgrounds from augmented reality, virtual reality, telecommunications, arts, and communications involved in industries or academia.

Gábor Sörös is a researcher in Nokia Bell Labs (Budapest, Hungary). His research interests span from wearable computer vision and spatial AI over augmented humans, augmented reality, and augmented intelligence in general, often with applications in interaction with robots and smart things. Gábor is a board member and technology lead in the Open AR Cloud EU association and is also engaged in standardization at ETSI and the Metaverse Standards Forum. Previously, he was an early contributor of 3 successful AR startups of ETH Zurich. Gábor holds a PhD from ETH Zurich in ubiquitous computing.

[2] https://www.openarcloud.org/
[3] https://www.ogc.org/
[4] https://metaverse-standards.org/
[5] https://immersal.com/
[6] https://augmented.city/
[7] https://xr-masters.com/

James Jackson is a Member of Technical Staff at AT&T Labs (Austin, TX, USA), Director of Technology and board member at Open AR Cloud, and co-chair of the Real/Virtual World Integration Working Group at the Metaverse Standards Forum. He has worked in telecom for the past 25 years building large-scale real-time distributed systems and promoting innovation. James is especially interested in applications of machine perception that improve society such as self-driving cars and an open interoperable Spatial Web. James received a BSEE from the University of Texas at Austin, and an MSCS (AI/ML focus) from Georgia Tech.

Mikel Salazar is a Human-Computer Interaction researcher focusing on the development of advanced 3D User Interfaces for Extended Reality systems. Currently, he is working at the Institute for Energy Technology (Halden, Norway). As the main developer of the GeoPose Sandbox, he writes a simple reference implementation and the associated documentation to enable others to translate them into different languages and build novel interaction techniques on top of them.

Christine Perey is an industry analyst, independent researcher and an outspoken advocate for interoperable Augmented Reality. From 2009-2016, she led the grassroots community dedicated to this purpose. She is a member of numerous standards development organizations and co-chairs the IEEE SA P2874 Spatial Web WG, the OGC Points of Interest SWG and the OGC GeoPose SWG. She is the founder of the AR for Enterprise Alliance (AREA), chairs the Research Committee, and leads the AREA Interoperability and Standards Program. Since 2018, she is a founder of the Open AR Cloud (OARC) association and serves on the governing board. Since 2022, she is a WG leader, member of the board of directors and an elected officer of the Metaverse Standards Forum.

Ali C. Hantal is a serial entrepreneur and passionate advocate for interoperability. He is the founder of XR Masters, Co-President of the Open AR Cloud Association, and Adoption Officer for the GeoPose Standards Working Group at the Open Geospatial Consortium (OGC). With a deep commitment to education and emerging technologies, he focuses on advancing XR and spatial computing through open standards and global collaboration. His company, XR Masters, is a U.S.-based XR technology firm behind MyGeoVerse—a platform capable of publishing spatial content in compliance with OGC and OSCP standards. Powered by AR Cloud technology and conversational AI avatars, MyGeoVerse bridges the physical and digital worlds to deliver immersive, context-aware experiences across tourism, education, and retail.

Alina Kadlubsky is a Reasearch Associate at the Department of Virtual Technologies at Applied Science University of Mittweida. She specializes in spatial computing and XR research, leveraging on creating a human-centered interoperable future of computing while contributing to various initiatives, publications and standards. She is a co-founding member of XRSI Europe and a founding board member of the SafeZone e.V. Alina has been leading the EU activities of Open AR Cloud Europe gUG as managing director through EU-funded research projects like NGI Atlantic, NGI Search and the Horizon-funded XR4Human.

Acknowledgments

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